

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7				Indicative content - ions are free to move because aluminium oxide is molten - ions travel to the oppositely charged electrodes because opposite charges attract / Al^{3+} ions go to the cathode / O^{2-} ions go to the anode - reduction is gain of electrons, oxidation is loss of electrons at the cathode – reduction $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ - aluminium ions gain 3 electrons to form aluminium atoms at the anode – oxidation $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ - two oxide ions lose 2 electrons each to form two oxygen atoms - these pair up to form oxygen molecules 5-6 marks Good explanation of the movement of ions towards both electrodes and of reduction and oxidation of both ions in terms of gain/loss of electrons; attempt at electrode equations <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		